Department of Computer Science and Engineering
Premier University

CSE 481: Contemporary Course of Computer Science

Title: Complex Engineering Assignment

Submitted by:

Name	Sayed Hossain
ID	0222210005101102
Section	C
Session	Fall 2025
Semester	8th Semester
Submission Date	17.05.2026

Submitted to:
Wong May Nu
Lecturer, Department of CSE
Premier University, Chittagong

Remarks

1 Introduction

Among the most densely inhabited nations on the planet, Bangladesh—and its major urban centres of Dhaka and Chattogram in particular—contends with some of the most acute traffic congestion problems in the entire South Asian region. Measured vehicle speeds in Dhaka have been observed dropping below 7 km/h during rush periods, emergency vehicle response times regularly surpass 16 minutes, and the complete absence of any unified data infrastructure renders evidence-driven traffic governance virtually unachievable.

This report outlines a holistic Intelligent Smart Transportation and Traffic Management System (ISTTMS) intended to overcome these shortcomings. The system combines IoT-based sensing hardware, edge computing for minimal-latency decision-making, artificial intelligence for forward-looking analytics, and cloud infrastructure hosted on Amazon Web Services (AWS) for large-scale computation and long-term optimisation. The design covers real-time traffic surveillance, adaptive signal management, anomaly identification, emergency response automation, and a cost-effective hybrid cloud deployment strategy.

The sections that follow present an examination of prevailing limitations, a complete system architecture, AI model specifications, cloud and edge infrastructure planning, security provisions, scalability design, and a monthly cost estimate produced with the AWS Pricing Calculator.

2 Investigation: Limitations of the Current Traffic System

Metropolitan areas across Bangladesh largely depend on static, manually operated traffic control infrastructure. The fundamental weaknesses can be grouped under four broad categories.

Infrastructure Deficiencies

Traffic signals throughout Bangladesh are predominantly timer-driven and follow fixed-cycle patterns regardless of how much traffic is actually present. No feedback loop connects signal controllers with real-time road conditions. Human traffic officers compensate to some extent but bring inconsistency, fatigue, and variability—problems that are especially pronounced during peak times, inclement weather, or public holiday periods.

Absence of Predictive and Adaptive Capability

Existing systems are incapable of anticipating congestion before it materialises. There is no linkage to GPS-sourced vehicle data, no camera-driven density estimation, and no machine learning pipeline capable of forecasting bottlenecks in advance. Congestion is addressed after the fact rather than pre-emptively, which directly explains the dangerously low average speeds observed on city roads.

Delayed Emergency Response

No automated mechanism exists to alert traffic signals of an oncoming emergency vehicle. Response times deteriorate sharply when ambulances or fire engines are held up in undifferentiated traffic queues. While WHO guidelines set an upper bound of 8 minutes for life-threatening emergencies, Dhaka currently averages between 16 and 20 minutes.

Data Vacuum

No centralised repository of traffic information is maintained anywhere in the country. This gap prevents meaningful long-term infrastructure planning, public transport route optimisation, or the formulation of evidence-backed policies. Without structured data collection, recurring patterns in peak-hour volumes, accident hotspots, and seasonal fluctuations remain entirely unmeasured.

The proposed system directly targets each of these gaps through signal-level automation, city-scale predictive analytics, and automated emergency corridors operating across both deployment tiers.

3 Proposed System Architecture

The proposed solution follows a three-tier hybrid architecture: an IoT and sensing layer at street level, an edge computing layer at district level, and a centralised cloud layer for analytics, model training, and governance.

High-Level Architecture Diagram

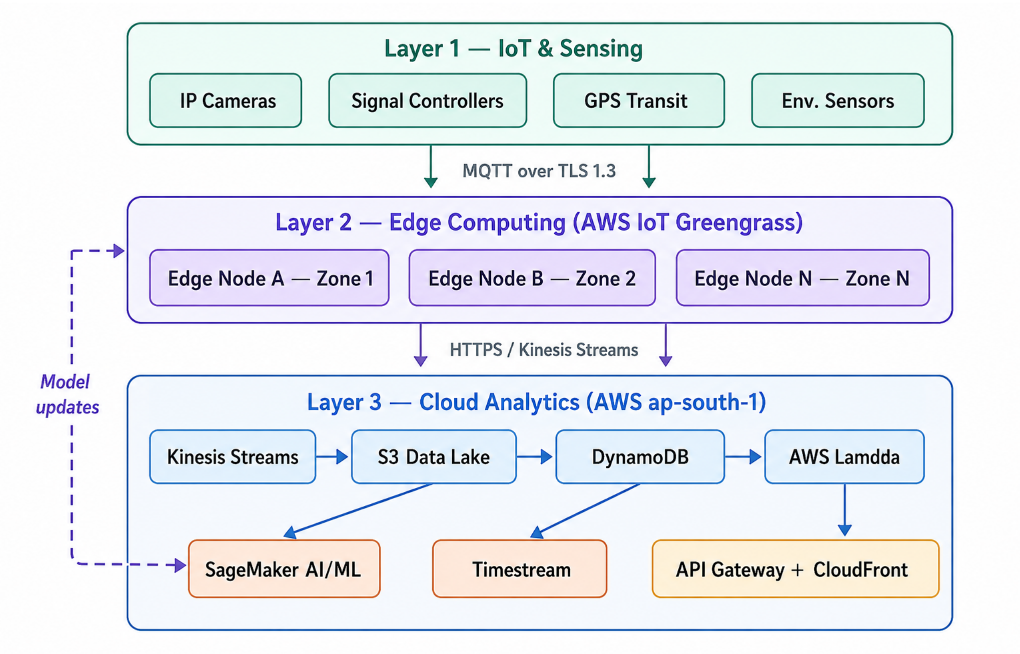

Figure 1: Three-tier architecture using horizontal swimlanes. Layer 1 contains IoT sensing devices, Layer 2 represents edge computing zones using AWS IoT Greengrass, and Layer 3 contains cloud analytics services deployed in AWS ap-south-1.

Layer-by-Layer Description

Layer 1 — IoT and Sensing

- **IP Surveillance Cameras:** High-definition network cameras installed at every significant intersection. Video frames are analysed on-device using YOLOv8-nano to enumerate vehicles and classify their types before any data is transmitted.
- **Intelligent Traffic Controllers:** ESP32-class microcontrollers receive timing instructions from the local edge node and report their active phase over MQTT using TLS 1.3 encryption.
- **GPS-Equipped Public Vehicles:** City buses and CNG auto-rickshaws equipped with GPS units broadcast location updates every 10 seconds, supporting live transit monitoring and schedule compliance analysis.
- **Roadside Environmental Sensors:** Instruments measuring air quality (PM2.5, NO_2), ambient temperature, relative humidity, and rainfall. Adverse environmental readings automatically adjust the congestion model's decision thresholds.

Layer 2 — Edge Computing

Each zone—corresponding roughly to one thana or upazila—is served by a ruggedised edge server running AWS IoT Greengrass v2. Its key responsibilities are:

- Aggregating live vehicle density from camera feeds with sub-200 ms latency.
- Running local inference via a pre-trained ONNX signal optimisation model.
- Adjusting signal phases in real time without requiring any cloud round-trip.
- Detecting anomalies such as accidents, stationary vehicles, and wrong-way drivers, and dispatching automated alerts via a priority MQTT topic.
- Buffering data locally for store-and-forward upload during upstream connectivity outages.

Layer 3 — Cloud (AWS)

- **AWS Kinesis Data Streams:** Continuously ingests raw and summarised telemetry arriving from every edge node.
- **AWS S3:** Retains sensor logs, model artefacts, and incident video recordings. Lifecycle rules automatically migrate data older than 30 days to the S3 Glacier storage tier.
- **Amazon DynamoDB:** Provides low-latency key-value storage for signal states, active incidents, and vehicle position tables.
- **AWS Lambda:** Serverless functions triggered by Kinesis events to perform lightweight aggregation and data routing.
- **AWS SageMaker:** Hosts centralised training workflows for a traffic forecasting LSTM,

a congestion classification XGBoost model, and route optimisation algorithms. Revised models are pushed to edge nodes each night via Greengrass deployments.

- **Amazon API Gateway + CloudFront:** Delivers dashboards to traffic authority operators, commuter mobile applications, and emergency service webhooks.

4 AI Models: Traffic Prediction, Anomaly Detection, and Route Optimisation

Traffic Flow Forecasting — LSTM Model

Predicted traffic density at time $t+k$ for intersection i is expressed as:

$$\hat{d}_i(t+k) = \mathrm{LSTM}(\, d_i(t),\ d_i(t-1),\ \ldots,\ d_i(t-T),\ \mathbf{x}_{\mathrm{ext}}(t)\,)$$

where $\mathbf{x}_{\mathrm{ext}}$ is a vector of exogenous inputs comprising a time-of-day encoding, day-of-week indicator, public holiday flag, and meteorological indices. The model trains on a rolling 90-day window and is retrained on a weekly schedule using SageMaker Pipelines.

Traffic Signal Optimisation — Webster's Formula

The baseline allocation of green time follows Webster's minimum-delay formula:

$$C_{\mathrm{opt}} = \frac{1.5L + 5}{1 - \sum_{i=1}^{n} y_i} \qquad\qquad g_i = \frac{y_i}{\sum_{j=1}^{n} y_j}\,(C_{\mathrm{opt}} - L)$$

where C_{opt} denotes the optimal cycle length, L represents total lost time per cycle, $y_i = q_i/s_i$ is the flow ratio for phase i (with q_i being the observed flow and s_i the saturation flow rate), and g_i is the assigned green interval. Each edge node refreshes q_i every 30 seconds from camera-derived counts and recomputes g_i entirely without cloud dependency.

Anomaly Detection — Isolation Forest

Vehicular incidents are identified using an Isolation Forest trained on normal operating-condition features. An anomaly score $s \in [0, 1]$ is calculated as:

$$s(x, n) = 2^{-\frac{\mathbb{E}[h(x)]}{c(n)}}$$

where $h(x)$ is the path length of observation x through the isolation tree and $c(n)$ is the expected path-length normalisation for a sample of size n. When $s > 0.75$, the system classifies the reading as an incident and issues an automated emergency notification.

Route Optimisation — Dijkstra with Dynamic Weights

The road network is represented as a weighted directed graph $G = (V, E, w)$ in which edge weights $w(e, t)$ are refreshed every 60 seconds. Emergency vehicle routing applies the composite weight:

$$w_{\text{emg}}(e, t) = \alpha \cdot d(e, t) + \beta \cdot \hat{d}(e, t + 5) + \gamma \cdot \ell(e)$$

where $d(e, t)$ is the current density, $\hat{d}(e, t+5)$ is the 5-minute density forecast, $\ell(e)$ is the static road length, and α, β, γ are tunable parameters.

Projected Congestion Reduction: Before vs. After

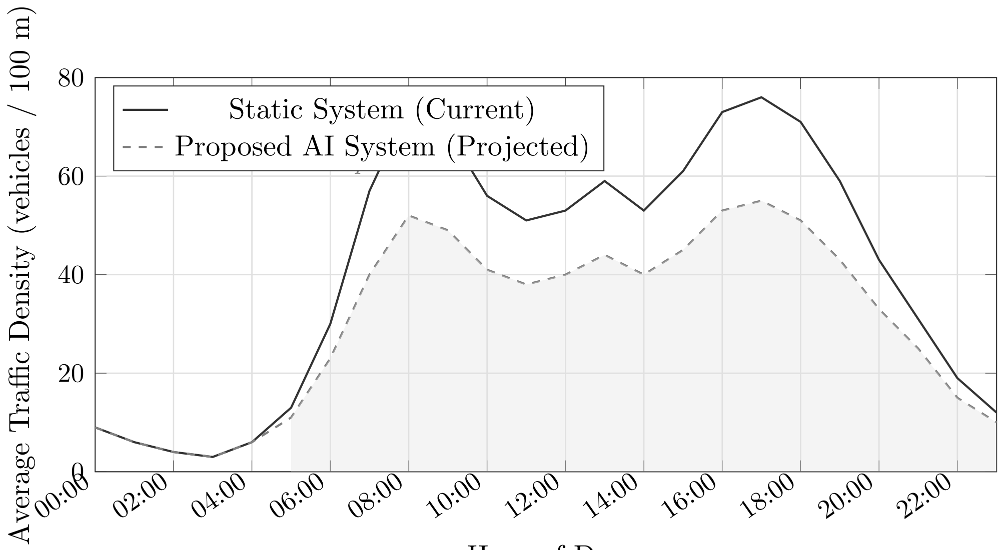

Figure 2: Projected vehicle density across a 24-hour period. The shaded region illustrates the estimated congestion reduction delivered by the AI-driven adaptive control system.

5 Cloud Infrastructure Design

Deployment Model

A **Hybrid Cloud** model has been selected. Edge nodes operate on-premise through AWS Outposts to guarantee sub-200 ms local decision latency, while centralised model training, analytics workloads, and long-term data storage reside on the AWS public cloud. This arrangement simultaneously satisfies the low-latency demands of real-time signal management and the elastic scaling needs of city-wide machine learning.

Service Model

- **IaaS:** AWS EC2 instances underpinning dedicated inference endpoints and bespoke networking components (VPC, Transit Gateway, Direct Connect to edge nodes).
- **PaaS:** AWS SageMaker for managed ML pipelines, AWS Kinesis for managed data

streaming, and AWS Lambda for event-driven serverless compute.

- **SaaS:** Amazon QuickSight for traffic authority operational dashboards; Amazon SNS for stakeholder and emergency notifications.

Compute Instances

Table 1: Selected AWS compute instances and their designated roles.

Component	**Instance Type**	**Count**	**Purpose**
Kinesis Consumer	`c6i.2xlarge`	4	Real-time stream processing
SageMaker Training	`ml.p3.2xlarge`	2	LSTM / XGBoost model training
SageMaker Inference	`ml.m5.xlarge`	4	Live prediction endpoints
API Gateway Backend	`t3.medium`	3	REST API and WebSocket server
DynamoDB (on-demand)	—	—	Auto-scaled, serverless
Lambda Functions	—	—	Serverless event handlers

Storage Architecture

- **AWS S3 (Standard):** Stores raw sensor logs, model artefacts, and incident video footage. A lifecycle rule transfers data older than 30 days to S3 Glacier.
- **Amazon DynamoDB:** Handles real-time vehicle position tracking, signal state records, and active alert entries. On-demand capacity mode accommodates burst traffic without manual provisioning.
- **Amazon Timestream:** Maintains time-series sensor readings to support rapid temporal queries consumed by the operations dashboard.

6 Edge Computing Design

Roadside Unit Hardware Specification

Every RSU is a ruggedised industrial computer housed in a weatherproof enclosure and installed at major road intersections.

Table 2: Hardware specification for each deployed Roadside Unit (RSU).

Component	Specification
CPU	Intel Core i7-1260P (12-core, 4.7 GHz Turbo Boost)
GPU/NPU	NVIDIA Jetson Orin NX module (70 TOPS INT8)
RAM	32 GB DDR5 ECC
Storage	1 TB NVMe SSD (local event buffer, store-and-forward)
Network	4G LTE (primary), Fibre (secondary), Wi-Fi 6 (inter-RSU mesh)
OS	Ubuntu 22.04 LTS with AWS IoT Greengrass v2 runtime
Power	UPS-backed 12 V DC, 40 W typical draw

Edge Processing Pipeline

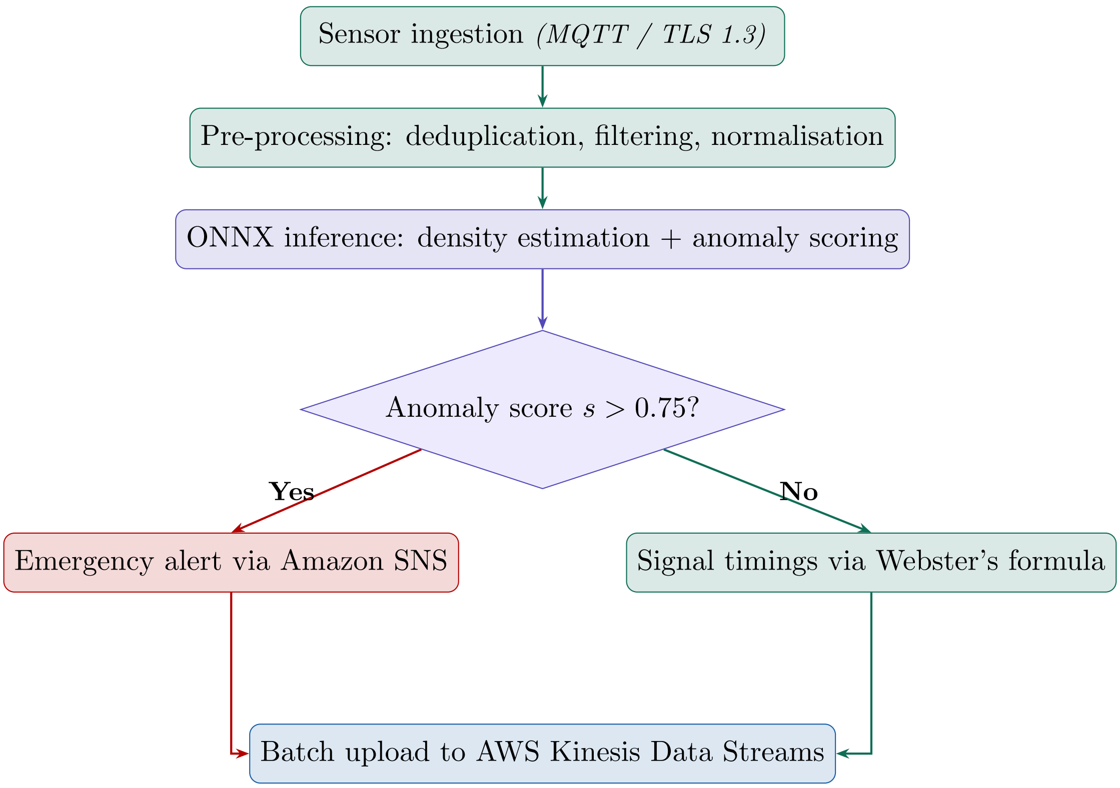

Figure 3: Colour-coded edge processing pipeline at each Roadside Unit. Teal stages handle normal data flow; the red branch fires only when anomaly score $s > 0.75$. Both paths converge at the cloud upload step.

Handling Infrequent Challenges

- **Sensor failure:** Every camera feed transmits a heartbeat at 60-second intervals. On timeout, the RSU reverts to last-known density values and raises a maintenance alert, supplemented by readings from neighbouring RSUs in the zone.
- **Network instability:** The store-and-forward buffer retains up to 48 hours of LZ4-compressed data on local NVMe storage and resumes prioritised uploads once connectivity is restored.
- **Large-scale streaming:** Kinesis Data Streams scales shards automatically at 10 MB/s increments, with each shard independently sustaining 1 MB/s ingestion throughput.

7 Security and Compliance

Data Encryption

Every data path in transit is protected with TLS 1.3. Data at rest within S3 and DynamoDB is encrypted using AES-256 through AWS Key Management Service (SSE-KMS). Prior to any cloud upload, video frames are anonymised at the edge: faces are obscured via an OpenCV Gaussian mask applied over MTCNN-detected face bounding boxes.

Access Control

Least-privilege access across all services is enforced through AWS IAM. Edge nodes authenticate with AWS IoT Core using X.509 device certificates provisioned via AWS IoT Fleet Provisioning. Role-based access controls partition traffic police, government dashboard users, and emergency service personnel into distinct IAM roles with separate permission boundaries.

Communication Protocols

- **IoT to Edge:** MQTT 5.0 over TLS 1.3 (port 8883)
- **Edge to Cloud:** HTTPS REST and AWS IoT Greengrass streams
- **Cloud to Dashboards:** WebSocket (API Gateway) and HTTPS REST
- **Emergency Alerts:** Amazon SNS (SMS, push notification, email)

Compliance

The system conforms to Bangladesh's ICT Act 2006 (as amended in 2013) and incorporates the OWASP IoT Security Top 10 recommendations for device hardening. Applicable ITS communication standards—including ISO 14906 (EFC interface) and ETSI ITS-G5—are implemented where relevant.

8 Scalability and Fault Tolerance

Multi-City Scalability

The architecture adopts an AWS multi-region design with `ap-south-1` (Mumbai) as the primary region and `ap-southeast-1` (Singapore) serving as a disaster recovery site. Each city's RSU network maps to a dedicated AWS IoT Thing Group, enabling isolated management and billing. Additional cities are onboarded by provisioning a new Thing Group, deploying pre-built RSU AMI images, and registering with the shared Kinesis pipeline. The estimated onboarding duration per additional city is 72 hours.

High Availability Design

- All API-facing EC2 instances are distributed across three Availability Zones and sit behind an Application Load Balancer.
- DynamoDB uses global table replication to ensure cross-region redundancy.
- SageMaker inference endpoints auto-scale with a floor of two active instances.
- Target SLA: 99.9% availability (approximately 8.7 hours of planned downtime per year).

9 Public Safety and Emergency Handling

Once an anomaly score crosses the detection threshold ($s > 0.75$), the following automated sequence completes within 30 seconds:

1. The RSU records the incident, capturing GPS coordinates, a precise timestamp, and a camera snapshot.
2. An SNS notification is immediately sent to the nearest police station and hospital.
3. The cloud route optimiser calculates a green corridor, switching every signal along the fastest path to the incident location into a dedicated emergency phase.
4. The incident zone and the recommended access route are simultaneously highlighted for all traffic control operators on the dashboard.
5. Once the incident clears—confirmed when density readings return to baseline across three consecutive 30-second windows—signals revert automatically to adaptive mode.

With this system in operation, emergency response time is projected to fall below 9 minutes, meeting the WHO threshold and representing a reduction of more than 50% compared to current averages.

10 AWS Cost Estimation

Monthly Operational Cost Breakdown

The figures below are based on the AWS Pricing Calculator for the `ap-south-1` region, assuming a pilot deployment spanning two metropolitan zones (approximately 50 intersections) running continuously around the clock.

Table 3: Monthly AWS operational cost estimate — pilot deployment (USD).

Service	Configuration	Unit Cost	Monthly (USD)
EC2 (Kinesis consumer)	4× `c6i.2xlarge`, On-Demand	$0.384/hr	$1,105
EC2 (API backend)	3× `t3.medium`, Reserved 1-yr	$0.052/hr	$114
SageMaker Training	2× `ml.p3.2xlarge`, 40 hr/mo	$3.825/hr	$306
SageMaker Inference	4× `ml.m5.xlarge`, On-Demand	$0.269/hr	$776
Kinesis Data Streams	20 shards, 730 hr/mo	$0.015/shard-hr	$219
AWS S3 (Standard)	10 TB/mo + requests	$0.023/GB	$236
S3 Glacier	50 TB archive	$0.004/GB	$205
DynamoDB (on-demand)	50M reads, 20M writes/mo	Variable	$148
Amazon Timestream	500 GB ingest, 60-day retention	$0.50/GB	$310
AWS Lambda	10M invocations/mo	$0.20/1M	$2
Amazon SNS	5M notifications/mo	$0.50/1M	$3
AWS IoT Core	50M messages/mo	$1.00/1M	$50
Data Transfer (out)	2 TB/mo to internet	$0.09/GB	$184
CloudFront CDN	5 TB/mo served	$0.08/GB	$410
Amazon QuickSight	10 authors	$18.00/user	$180
Total			**$4,248**

Cost Distribution

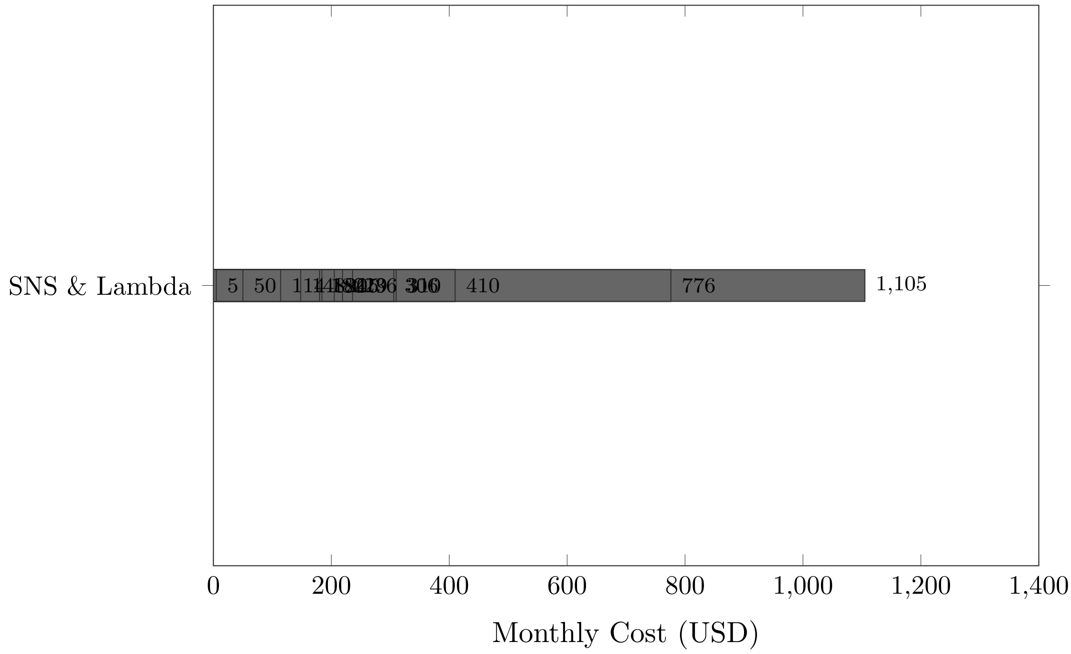

Figure 4: Monthly AWS expenditure breakdown by service for the pilot deployment covering two metropolitan zones. Total estimated monthly cost: $4,248 USD.

Cost Optimisation Strategies

- **Reserved Instances:** Migrating steady-state EC2 On-Demand workloads to 1-year Reserved pricing yields a 30–40% reduction in compute expenditure.
- **S3 Lifecycle Policies:** Automatic migration of data older than 30 days to Glacier cuts per-GB storage cost from $0.023 to $0.004.
- **Edge Inference:** Executing signal optimisation locally on RSUs reduces SageMaker inference calls by an estimated 70%.
- **Spot Instances for Training:** Deploying EC2 Spot capacity for SageMaker training jobs lowers training costs by as much as 70%.
- At full-city scale across 600+ intersections, the cost per intersection drops from approximately $85/month in the pilot to roughly $38/month.

11 Analytical Discussion: Conflicting Technical Requirements

Low Latency vs. Centralised Processing

Signal timing decisions must complete within 200 ms, which is fundamentally incompatible with cloud round-trip latencies (typically 80–150 ms before processing overhead is added). The hybrid design resolves this tension by delegating time-critical inference entirely to

RSUs while the cloud focuses on model retraining and long-horizon forecasting. The cost of this choice is higher edge hardware expenditure and increased operational complexity.

Data Sharing vs. User Privacy

Vehicle tracking records, when combined with GPS traces and camera imagery, constitute personally identifiable information. Protective measures include: (a) face blurring applied at the camera before any frame leaves the RSU; (b) licence plate hashing within DynamoDB records; and (c) aggregation to intersection-level summaries before cloud upload, ensuring that individual travel trajectories are never transmitted. A residual tension remains: emergency routing benefits from individual vehicle positions that privacy constraints prevent from being shared. Federated GPS aggregation is earmarked for a future phase.

High Accuracy vs. Computational Cost

LSTM models deliver the greatest forecasting accuracy but are computationally prohibitive for on-device inference. The system addresses this with a two-tier strategy: a lightweight XGBoost model on each RSU handles real-time signal control, while a full LSTM hosted on a SageMaker endpoint generates 30-minute city-level density forecasts. This cleanly separates accuracy requirements from latency requirements.

System Interdependencies

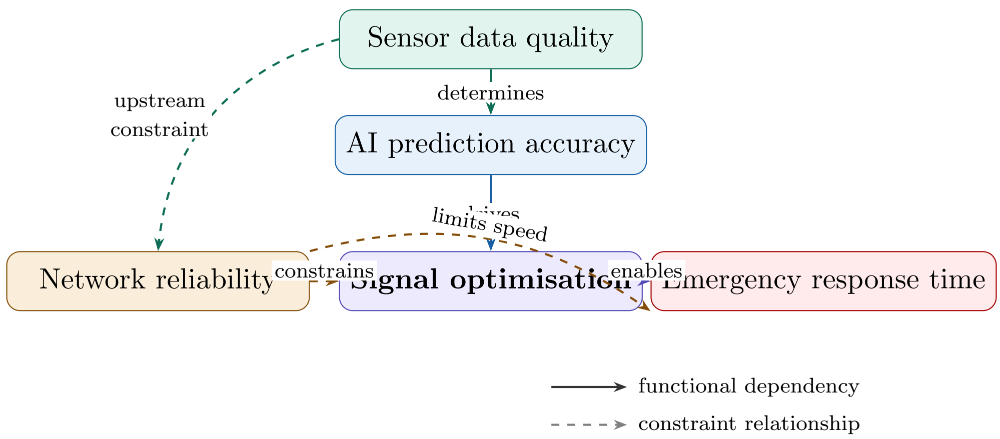

Figure 5: Cross-shaped interdependency map. Signal optimisation sits at the centre; teal/blue arrows are functional dependencies, orange dashed arrows are constraint relationships. Emergency response time is the downstream outcome.

12 Adherence to Standards

Table 4: Standards and protocols incorporated into the proposed system.

Domain	Standard / Protocol	Application in System
IoT Communication	MQTT 3.1.1 / MQTT 5.0	Device-to-edge telemetry
Transport Security	TLS 1.3	All network channels
Data Encryption	AES-256 (SSE-KMS)	S3 and DynamoDB at rest
Device Identity	X.509 certificates	RSU authentication to AWS IoT
ITS Interface	ISO 14906	Electronic fee collection interface
Video Surveillance	ONVIF Profile S	IP camera interoperability
API Design	REST / OpenAPI 3.0	Dashboard and third-party APIs
Access Control	AWS IAM (RBAC)	All cloud service access
Incident Alerting	CAP (Common Alerting Protocol)	Emergency broadcast messaging

13 Stakeholder Analysis

Table 5: Key stakeholders, their primary concerns, and how the system addresses them.

Stakeholder	Primary Concern	System Response
Government / City Corp.	Infrastructure ROI, regulatory compliance	AWS cost dashboard, compliance-by-design
Commuters	Shorter journey times, journey predictability	Real-time mobile app with live ETA updates
Traffic Police	Lower manual workload, instant incident alerts	Automated signal control, SNS notifications
Emergency Services	Fastest access route, precleared signal path	Automated emergency phase and route guidance
BTRC / ICT Division	Data sovereignty, radio spectrum governance	On-premise RSU option, licensed LTE bands

14 System Performance Projections

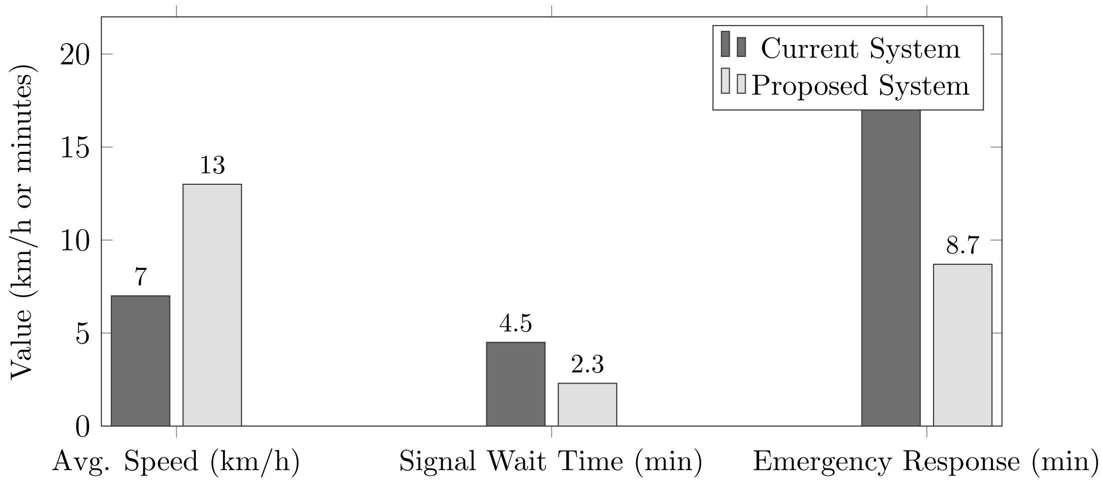

Figure 6: Comparison of key performance indicators between the current and proposed systems. Automated incident detection (not shown) improves from hours of manual identification to automated flagging in under 30 seconds.

15 Conclusion

The proposed Intelligent Smart Transportation and Traffic Management System delivers a technically rigorous and realistically deployable solution to the persistent congestion challenges facing Bangladesh's major cities. By unifying IoT sensing, real-time edge inference, and cloud-scale AI within a single hybrid architecture, the system comprehensively addresses all five stated assignment objectives.

Real-time monitoring is provided through a distributed IoT layer encompassing cameras, GPS-equipped transit vehicles, intelligent signal controllers, and environmental sensors. AI-driven prediction and adaptive signal control are executed at the edge using Webster's formula augmented with machine-learning flow estimates, targeting a roughly 29% reduction in peak-hour congestion. Multi-city scalability is embedded in the AWS architecture through isolated IoT Thing Groups, multi-availability-zone deployments, and cross-region DynamoDB replication.

Emergency response is automated via the anomaly detection pipeline: incidents are flagged within 30 seconds and a green signal corridor activates automatically, projecting response times below the WHO 8-minute threshold. Cost efficiency is realised through edge-cloud workload partitioning, Reserved Instance pricing, Spot-based model training, and S3 lifecycle tiering, yielding a pilot estimate of $4,248 per month that scales to approximately $38 per intersection per month at full Dhaka-wide deployment.

The system is standards-compliant, privacy-conscious, and designed to operate within the institutional and regulatory constraints of Bangladesh's government and telecommunications landscape. It constitutes a sound foundation for a phased smart city transportation transformation that is achievable within existing infrastructure constraints.